

CEAL

CIVIC EMERGENCY ACCESS LAYER

OmShantyOm



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The Problem: (



Public Systems Assume Connectivity

Cellular networks fail during disasters

Citizens cannot report emergencies

Rural areas face unreliable internet

Response times go untracked

Grievances go unfilled

Impact ?

Delayed emergency response, Lack of accountability, & Digital exclusion





Why Current Solutions Fail ?

“All current systems rely on telecom infrastructure.
If telecom fails, access to emergency services fails ”

Current System

Emergency call numbers

Safety apps

Location sharing

Government reporting portals

SMS emergency systems

Why it fails

Requires signal

Need mobile data

Internet dependent

Online only

Still require network





India's rural
teledensity lags
urban at 58.8% vs
125.3% (Q1 2025),
underscoring
offline needs in
disasters



Only 31% rural
households have
internet access vs
67% urban (2025),
fueling under-
reporting in crises



Why It Matters



India leads
globally with
440M offline
(2025), amplifying
last-mile reporting
barriers vs other
nations

Our Solution

A citizen-powered BLE mesh network that -



Broadcasts emergency & civic reports offline



Relays across nearby devices

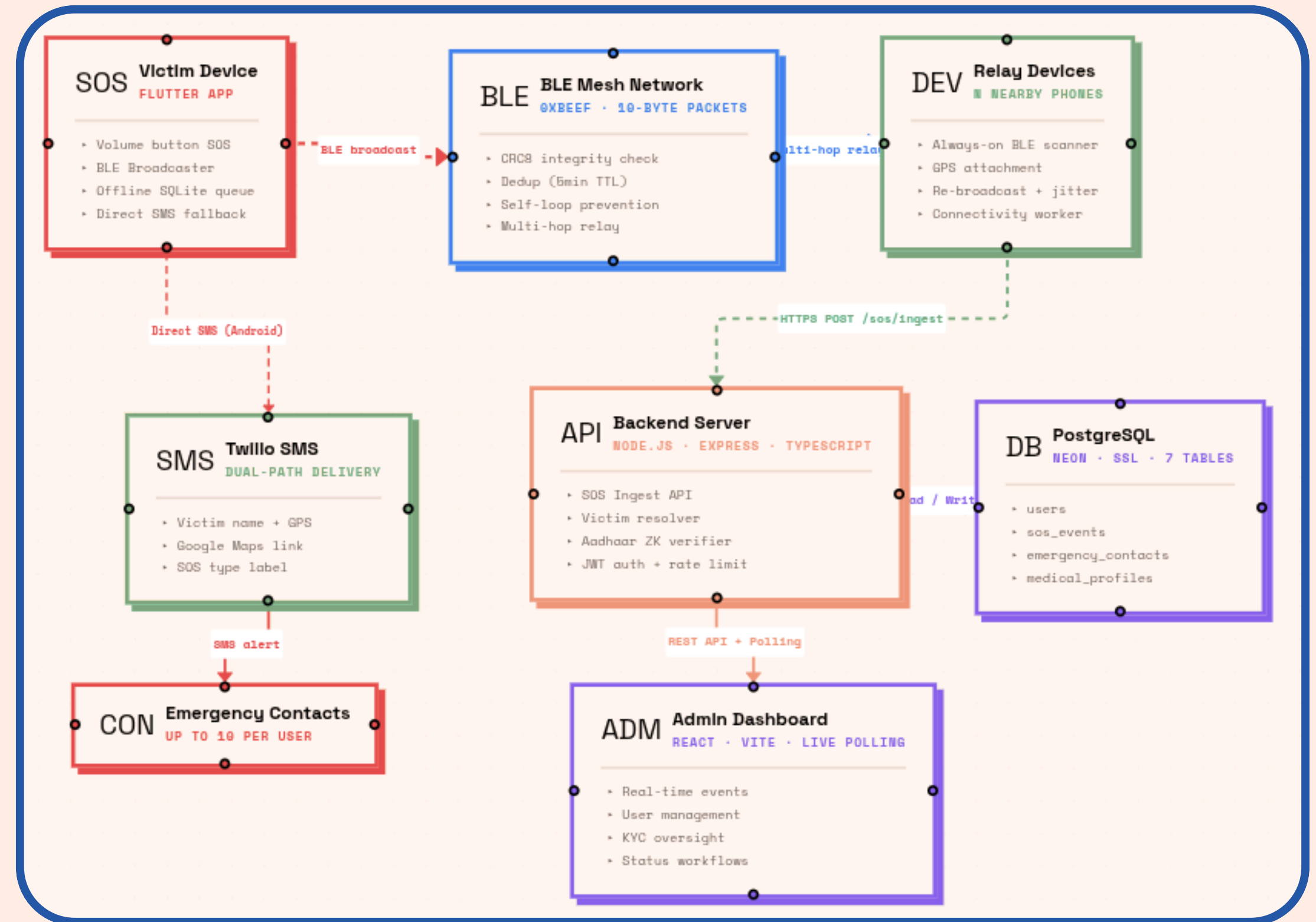


Escalates automatically when internet appears



Creates a tamper-evident incident log

How it works ?!?



Offline Mesh Propagation

Uses Bluetooth
Low Energy (BLE)

Short-range
multi-hop relay



TTL + hop control

Low battery
background mode

Duplicate
suppression

Why so low?

- Peer-to-peer relay reduces server load
- Data transmitted only during incidents
- No continuous streaming
- No expensive AI inference



Cloud hosting (shared infra)
~ ₹ 2.00 / person



Database storage
₹ 0.75 / person ~

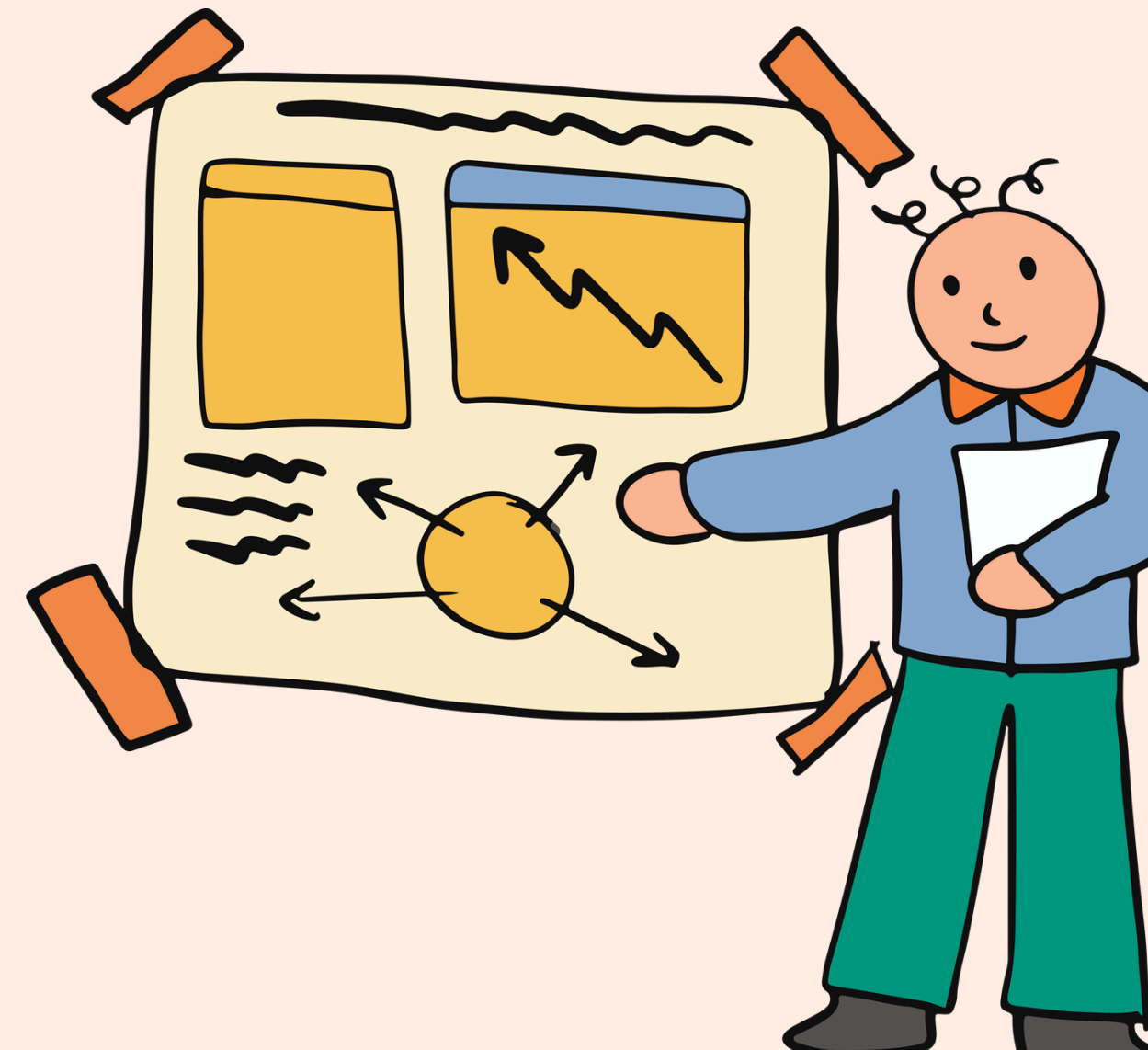
SMS fallback (rare use)
~ ₹ 0.50 / person

Monitoring / logs
~ ₹ 0.30 / person



Unit Economics

Unit



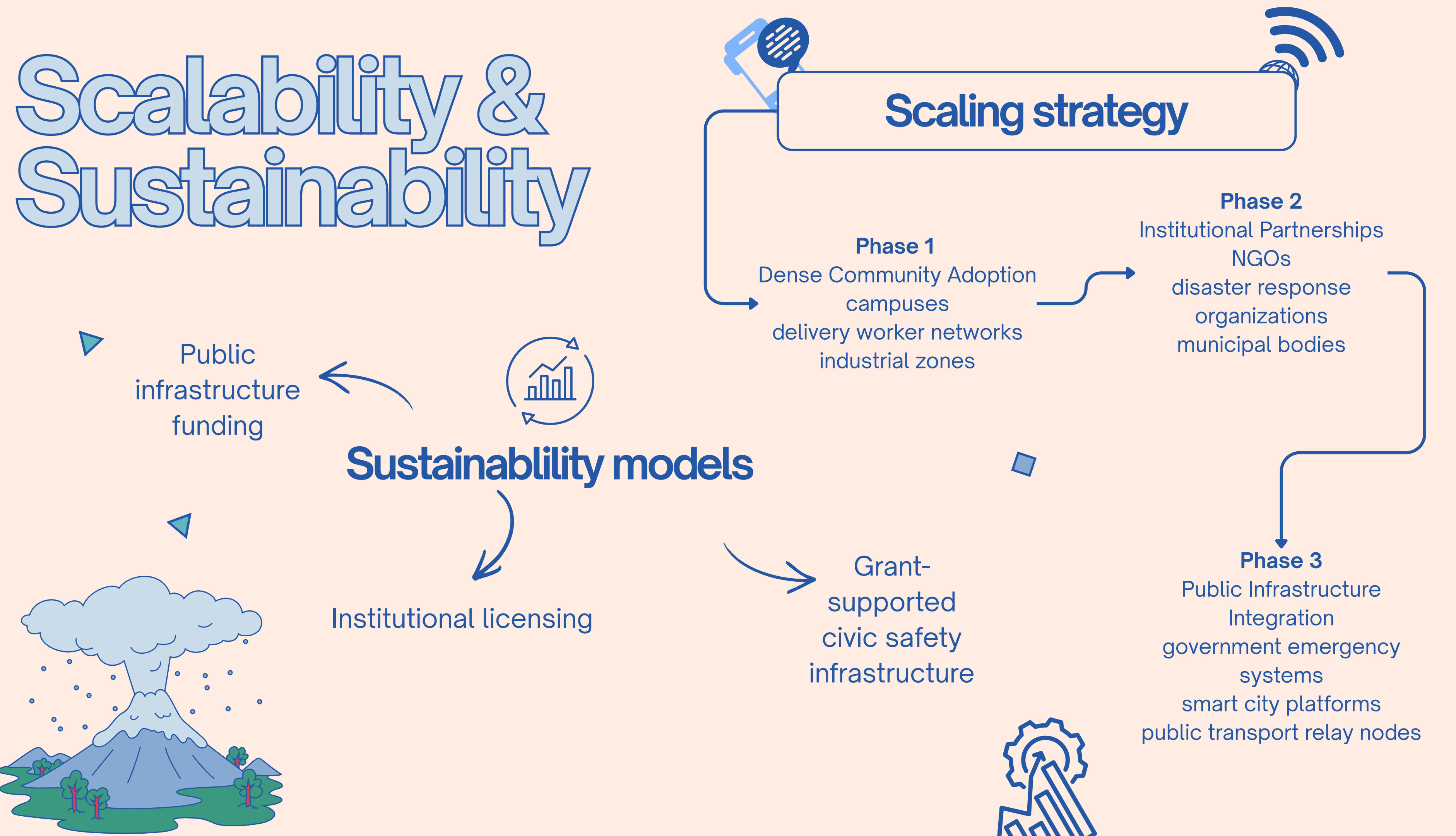
Not just SOS

- ✓ Infrastructure hazard reports
- ✓ Disaster observation
- ✓ Missing person alerts
- ✓ Public service complaints
- ✓ Local first-response signaling
- ✓ Admin Dashboard

**“This is not just safety.
This is last-mile governance
infrastructure”**



Scalability & Sustainability



The Ask



Funding

Substantial money investment for the project to sustain the early years



Partnerships

A network of countless stakeholders and organisation all working towards improving disaster response



Access

Permission and ability to run real-world pilot in one high-density zone

